

Geometric and algebraic reduction of multisymplectic manifolds

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Outline

- 1 Symplectic and multisymplectic manifolds
- 2 The L_∞ -algebra of observables
- 3 Symplectic reduction
- 4 Multisymplectic reduction
- 5 L_∞ -reduction

Section 1

Symplectic and multisymplectic manifolds

Symplectic manifolds

Fix a smooth manifold M .

Definition

A *symplectic structure* on M is a 2-form $\omega \in \Omega^2(M)$ which is

- i. *nondegenerate*: $\forall X \in TM \setminus \{0\} : \iota_X \omega \neq 0$,

$$TM \xrightarrow{\sim} T^*M$$

- ii. *closed*: $d\omega = 0$.

Theorem (Darboux)

There exist local coordinates $(x_1, \dots, x_n, y_1, \dots, y_n)$ in which

$$\omega = dx_1 \wedge dy_1 + \dots + dx_n \wedge dy_n.$$

Examples

Example

The standard symplectic structure on the smooth manifold $\mathbb{R}^{2n}$ is

$$\omega = dx_1 \wedge dy_1 + \cdots + dx_n \wedge dy_n,$$

where $\{x_1, \dots, x_n, y_1, \dots, y_n\}$ are standard coordinates.

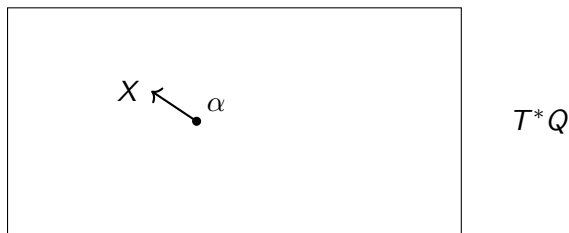
Example (cotangent bundles)

Let Q be a smooth manifold. The *canonical 1-form* $\theta \in \Omega^1(T^*Q)$ is given by

$$\theta_\alpha(X) = \alpha(\pi_*X)$$

and the *canonical symplectic structure* on T^*Q is defined to be $-d\theta$.

Cotangent bundle



$$\theta_\alpha(X) = \alpha(\pi_* X)$$



k -plectic manifolds

Fix a smooth manifold M .

Definition

A k -plectic structure on M is a $(k+1)$ -form $\omega \in \Omega^{k+1}(M)$ which is

- ① *nondegenerate*: $\forall X \in TM \setminus \{0\} : \iota_X \omega \neq 0$,

$$TM \hookrightarrow \Lambda^k T^*M$$

- ② *closed*: $d\omega = 0$.

There is **no local isomorphism property**.

Examples

Example (symplectic powers)

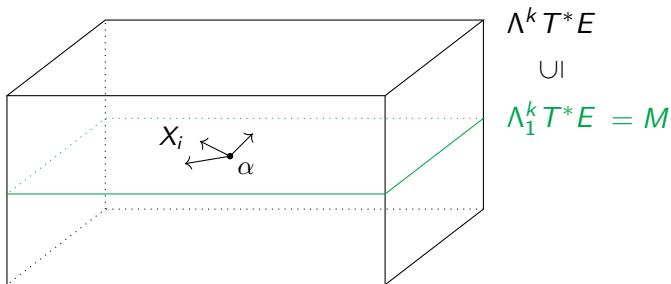
If (M^{2n}, σ) is a symplectic manifold, then σ^ℓ is a $(2\ell - 1)$ -plectic structure on M for $1 \leq \ell \leq n$.

Example (multiphase spaces)

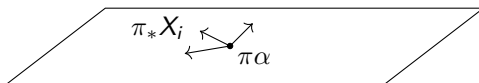
Let $\pi_E : E \rightarrow \Sigma$ be a smooth fiber bundle. We say that $\alpha \in \Lambda^k T_x^* E$ is *(1-)semihorizontal* if $\iota_u \iota_v \alpha = 0$ for all vertical tangent vectors $u, v \in \ker d(\pi_E)_x$, and denote by $\pi : \Lambda_1^k T^* E \rightarrow E$ the bundle of semihorizontal k -forms on E . The *canonical k -form* $\theta \in \Omega^k(\Lambda_1^k T^* E)$ is given by

$$\theta_\alpha(X_1, \dots, X_k) = \alpha(\pi_* X_1, \dots, \pi_* X_k), \quad X_i \in T_\alpha(\Lambda_1^k T^* E),$$

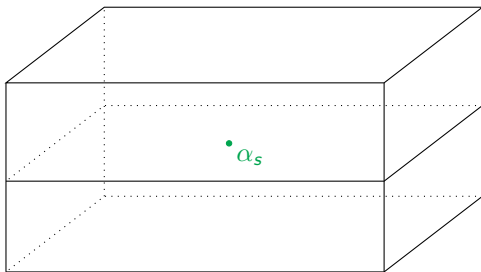
and the *canonical k -plectic structure* on $\Lambda_1^k T^* E$ is defined to be $-\mathrm{d}\theta \in \Omega^{k+1}(\Lambda_1^k T^* E)$.



$$\theta_\alpha(X_1, \dots, X_k) = \alpha(\pi_* X_1, \dots, \pi_* X_k)$$



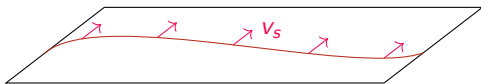
$$\begin{array}{c}
 E \\
 \downarrow \\
 \Sigma^{k-1}
 \end{array}$$



$$\begin{aligned} \Lambda^k T^*E \\ \cup \\ \Lambda_1^k T^*E = M \end{aligned}$$

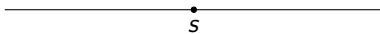
$$v \in T_\sigma \Gamma(E)$$

$$\alpha \in T_\sigma^* \Gamma(E)$$



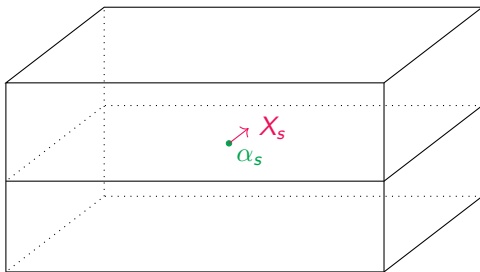
$$\sigma \in \Gamma(E)$$

$$\langle \alpha, v \rangle = \int_\Sigma \iota_v \alpha$$



$$\begin{aligned} E \\ \downarrow \\ \Sigma^{k-1} \end{aligned}$$

$$X \in T_\sigma(T^*\Gamma(E))$$



$$\Lambda^k T^*E$$

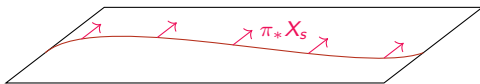
$$\cup$$

$$\Lambda_1^k T^*E = M$$

$$\pi_* X \in T_\sigma \Gamma(E)$$

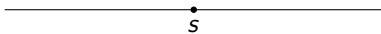
$$\alpha \in T_\sigma^* \Gamma(E)$$

$$\hat{\theta}_\alpha(X) = \int_\Sigma \iota_{X_s} \theta_{\alpha_s}$$



$$\left(\sigma \in \Gamma(E) \right)$$

$$\langle \alpha, \pi_* X \rangle = \int_\Sigma \iota_{\pi_* X} \alpha$$



$$E$$


$$\Sigma^{k-1}$$

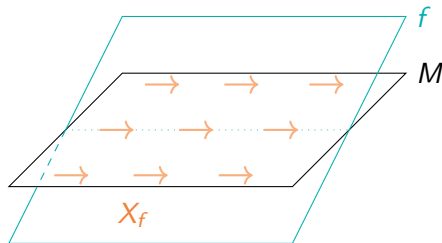
Section 2

The L_∞ -algebra of observables

The Poisson bracket on a symplectic manifold

observables $\longrightarrow$ symmetries
 $C^\infty(M) \ni f$ $X \in \mathfrak{X}(M)$, $\mathcal{L}_X \omega = 0$

$$df = -\iota_{X_f} \omega$$



Poisson bracket on $C^\infty(M)$:

$$\{f, h\} = \omega(X_f, X_h)$$

A multisymplectic analogue

$$\begin{array}{ccc}
 \text{observables} & \longrightarrow & \text{symmetries} \\
 \cancel{C^\infty(M)} \ni \cancel{f} & & X \in \mathfrak{X}(M), \quad \mathcal{L}_X \omega = 0 \\
 \Omega_{\text{ham}}^{k-1}(M) \ni \alpha & & \\
 d\alpha = -\iota_{X_\alpha} \omega & &
 \end{array}$$

Lie bracket on $\Omega_{\text{ham}}^{k-1}(M)$:

$$\{\alpha, \beta\} = \omega(X_\alpha, X_\beta, \cdot, \dots)$$

$$\{\alpha, \{\beta, \gamma\}\} + \{\gamma, \{\alpha, \beta\}\} + \{\beta, \{\gamma, \alpha\}\} = d\iota_{X_\alpha} \iota_{X_\beta} \iota_{X_\gamma} \omega$$

How to proceed?



Interesting alternative:

- Incorporate the **discrepancy** as part of the data of the space of observables

$$L_{\infty}(M, \omega) = \Omega_{\text{ham}}^{k-1}(M) \oplus \textcolor{red}{\Omega}^{\leq k-2}(M),$$

a **homotopy Lie algebra** or L_{∞} -algebra.

Informally, an ∞ -category is

- a category $\mathcal{C}$ with a **space of morphisms** $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ between each pair of objects $X, Y \in \mathcal{C}$,
- so that the usual categorical identities are required to hold only **up to homotopy**,
- and we **incorporate these (higher) homotopies** in the structure of $\mathcal{C}$.

keep track of
homotopies

∞ -category $\mathcal{C}$

quotient out by
homotopies

homotopy category $\mathrm{h}\mathcal{C}$

Higher categories appear in the L_{∞} -algebra of observables on a multisymplectic manifold.

The L_∞ -algebra of observables

$$L_\infty(M, \omega) := \Omega_{\text{ham}}^{k-1}(M, \omega) \oplus \Omega^{\leq k-2}(M)$$

with *higher brackets*

$$l_i : \underbrace{L_\infty(M, \omega) \otimes \cdots \otimes L_\infty(M, \omega)}_{i \text{ times}} \rightarrow L_\infty(M, \omega)$$

$$l_1(\alpha) = d\alpha$$

$$\begin{aligned} l_i(\alpha_1, \dots, \alpha_i) &= \varsigma(i) \iota_{X_1 \wedge \cdots \wedge X_i} \omega, & i \geq 2, \\ &= \pm \omega(X_1, X_2, \dots, X_i, \dots) \end{aligned}$$

where $\varsigma(i) = -(-1)^{\frac{i(i+1)}{2}}$.

$$\mathrm{Ham}_\infty(M, \omega) := \underbrace{\tilde{\Omega}_{\mathrm{ham}}^{k-1}(M, \omega)}_{\{(X_\alpha, \alpha)\}} \oplus \Omega^{\leq k-2}(M)$$

We define

$$\tilde{l}_i : \underbrace{\mathrm{Ham}_\infty(M, \omega) \times \cdots \times \mathrm{Ham}_\infty(M, \omega)}_{i \text{ times}} \rightarrow \mathrm{Ham}_\infty(M, \omega)$$

where

$$\tilde{l}_1(\alpha) = \begin{cases} (0, d\alpha) & \text{if } \alpha \in \Omega^{k-2} \\ d\alpha & \text{if } \alpha \in \Omega^{<k-2} \end{cases}$$

$$\tilde{l}_2((X_1, \alpha_1), (X_2, \alpha_2)) = ([X_1, X_2], \iota_{X_1 \wedge X_2} \omega)$$

$$\tilde{l}_i((X_1, \alpha_1), \dots, (X_i, \alpha_i)) = \varsigma(i) \iota_{X_1 \wedge \cdots \wedge X_i} \omega, \quad i \geq 3.$$

Section 3

Symplectic reduction

Conventions

- M and G are connected,
- $\xi, \zeta \in \mathfrak{g}$,
- $\lambda, \tau \in \mathfrak{g}^*$
- for $\mu \in \Omega^*(M, \mathfrak{g}^*)$ and $\xi \in \mathfrak{g}$, write

$$\mu_\xi := \langle \mu, \xi \rangle \in \Omega^*(M)$$

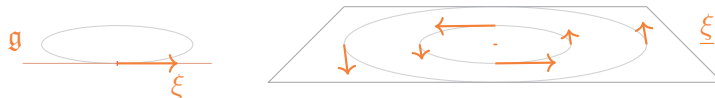
for the “ ξ th component” of μ .

Symplectic Hamiltonian actions

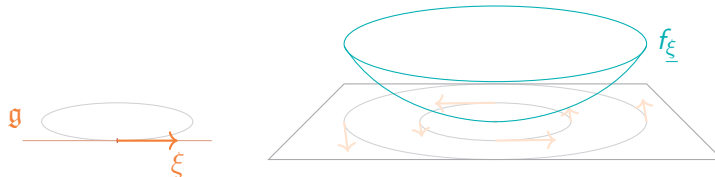
To specify a symplectic action $G \curvearrowright M \dots$



we could describe the induced map $\xi \mapsto \underline{\xi} \dots$



or an assignment of Hamiltonian functions $\xi \mapsto f_{\underline{\xi}}$.

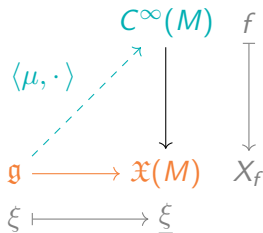


When this is possible,* the action is called **Hamiltonian**.

*and $\xi \mapsto f_{\underline{\xi}}$ is a homomorphism of Lie algebras

Symplectic Hamiltonian G -spaces

idea: describe $G \curvearrowright M$ in terms of $C^\infty(M) \rightarrow \mathfrak{X}(M)$



as Lie algebras

i.e., $\langle \mu, \xi \rangle$ generates $\underline{\xi}$

moment map: $\mu : M \rightarrow \mathfrak{g}^*$

Hamiltonian G -space: (M, ω, G, μ)

Symplectic reduction

Two inputs:

- 1 Hamiltonian G -space (M, ω, G, μ)
- 2 parameter $\lambda \in \mathfrak{g}^*$

The *reduced space* is $M_\lambda := \mu^{-1}(\lambda)/G_\lambda$.

Theorem (Marsden–Weinstein '74, Meyer '73)

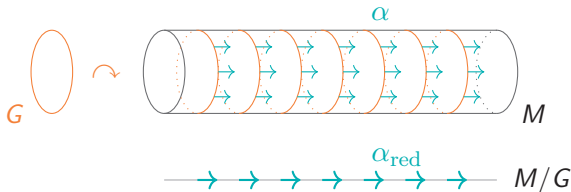
If $\mu^{-1}(\lambda) \subseteq M$ is smooth and $G_\lambda \curvearrowright \mu^{-1}(\lambda)$ is free and proper, then there is a unique symplectic form $\omega_\lambda \in \Omega^2(M_\lambda)$ such that $\pi^*\omega_\lambda = i^*\omega$.

$$\begin{array}{ccc} \text{restrict to } \{\mu = \lambda\} & \pi^*\omega_\lambda & \begin{array}{c} i^*\omega \\ \mu^{-1}(\lambda) \hookrightarrow M \end{array} \\ \text{quotient by } G_\lambda & \omega_\lambda & \begin{array}{c} \downarrow \pi \\ M_\lambda \end{array} \end{array}$$

The action descent lemma

If

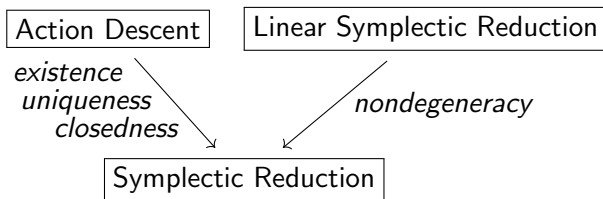
- $G \curvearrowright M$ free and proper,
- $\alpha \in \Omega^*(M)$ invariant and horizontal ($\iota_{\underline{g}}\alpha = 0$),



then

- $\exists! \alpha_{\text{red}} \in \Omega^*(M/G)$ such that $\alpha = \pi^* \alpha_{\text{red}}$,
- $d\alpha = 0 \implies d\alpha_{\text{red}} = 0$.

Symplectic reduction — proof outline



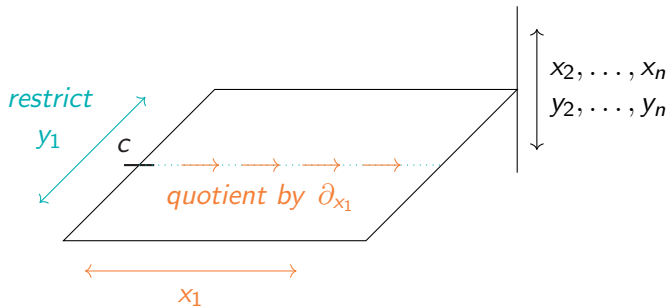
- ① Apply the *Action Descent Lemma* to $G_\lambda \curvearrowright \mu^{-1}(\lambda)$ and $i^*\omega$.

$$\begin{array}{ccc} i^*\omega & \mu^{-1}(\lambda) & \\ & \pi \downarrow & \\ \omega_\lambda & M_\lambda & \end{array}$$

- ② Use *Linear Symplectic Reduction* to conclude that ω_λ is nondegenerate.

Symplectic reduction — idea

$$\omega = \underbrace{dx_1 \wedge dy_1}_{\text{to be removed}} + dx_2 \wedge dy_2 + \cdots + dx_n \wedge dy_n$$



Restrict and quotient conjugate degrees of freedom.

Section 4

Multisymplectic reduction

Heuristic approach to reduction

1. describe $G \curvearrowright M$ in terms of ω
moment map μ
2. identify a distinguished reduced space
reduction at $\mu = 0$
3. use the ambiguity in 1. to obtain a family of reduced spaces
reduction at $\mu - \lambda = 0$,
i.e. reduction at $\mu = \lambda$

Note: If $G_\lambda \neq G$, then $\mu - \lambda : M \rightarrow \mathfrak{g}^*$ is not a moment map for either $G \curvearrowright M$ or $G_\lambda \curvearrowright M$.

A Leibniz bracket on $\Omega_{\text{ham}}^{k-1}(M)$

- ① $\{\alpha, \beta\} = \mathcal{L}_{X_\alpha}\beta \quad \leftarrow \text{Leibniz bracket (Jacobi, but not antisymmetric)}$
- ② $\{\alpha, \beta\}' = \iota_{X_\alpha} d\beta = -\omega(X_\alpha, X_\beta, \dots)$

Equal up to an exact term:

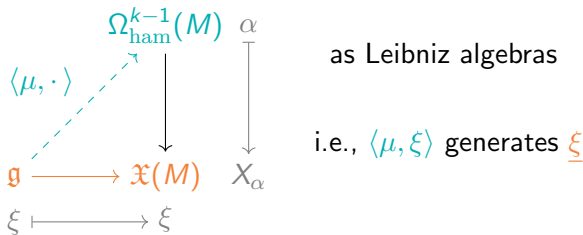
$$\mathcal{L}_{X_\alpha}\beta = \iota_{X_\alpha} d\beta + d\iota_{X_\alpha}\beta$$

$\alpha \mapsto X_\alpha$ is a homomorphism of Leibniz algebras:

$$\begin{aligned} d\{\alpha, \beta\} &= d\mathcal{L}_{X_\alpha}\beta = \mathcal{L}_{X_\alpha}\iota_{X_\beta}\omega = \iota_{[X_\alpha, X_\beta]}\omega \\ \implies X_{\{\alpha, \beta\}} &= [X_\alpha, X_\beta] \end{aligned}$$

Multisymplectic Hamiltonian G -spaces

idea: Describe $G \curvearrowright M$ in terms of $\frac{C^\infty(M)}{\Omega_{\text{ham}}^{k-1}(M)} \rightarrow \mathfrak{X}(M)$.



moment map: $\mu \in \Omega^{k-1}(M, \mathfrak{g}^*)$

Hamiltonian G-space: (M, ω, G, μ)

The moment map conditions

- $G \curvearrowright M$
- $\mu \in \Omega^{k-1}(M, \mathfrak{g}^*)$

1 Hamiltonian condition:

$$d\mu_\xi = \iota_\xi \omega$$

2 Leibniz condition:

$$\mu_{[\xi, \zeta]} = \{\mu_\xi, \mu_\zeta\}$$

equivalently,

$$\mathcal{L}_\xi \mu_\zeta = \mu_{[\xi, \zeta]}$$

The space of moment maps

- (M, ω, G, μ) Hamiltonian G -space
- $\phi \in \Omega^{k-1}(M, \mathfrak{g}^*)$

Question: When is $\mu + \phi$ a moment map?

- $d\phi = 0$, since

$$d(\mu + \phi)_\xi = \iota_\xi \omega \implies d\phi_\xi = 0.$$

- $\mathcal{L}_\xi \phi_\zeta = \phi_{[\xi, \zeta]}$, as

$$\mathcal{L}_\xi(\mu + \phi) = (\mu + \phi)_{[\xi, \zeta]} \implies \mathcal{L}_\xi \phi = \phi_{[\xi, \zeta]}.$$

i.e. ϕ is a moment map for the trivial action $G \curvearrowright M$.

The space of moment maps is an affine space modeled on $\{\phi \in \Omega^{k-1}(M, \mathfrak{g}^*) \mid d\phi = 0, G_\phi = G\}$.

Level sets of the moment map

Rather than:

- family of moment maps $\{\mu - \phi \mid d\phi = 0, G_\phi = G\}$
- reduction at $\mu - \phi = 0$

We instead consider:

- fixed moment map μ
- family of levels $\{\phi \mid d\phi = 0, G_\phi = G\}$
- reduction at $\mu = \phi$

ϕ -level set:

$$\mu^{-1}(\phi) := \{x \in M \mid \mu_x = \phi_x\}$$

Multisymplectic reduction

Two inputs:

- 1 Hamiltonian G -space (M, ω, G, μ)
- 2 parameter $\phi \in \Omega^{k-1}(M, \mathfrak{g}^*)$ with $d\phi = 0$

Define the *reduced space*,

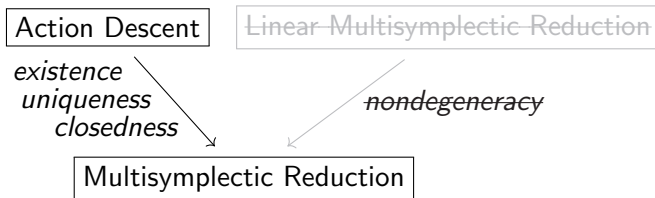
- $M_\phi = \mu^{-1}(\phi) / G_\phi$

Theorem (B. '21)

If $\mu^{-1}(\phi) \subseteq M$ is smooth and $G_\phi \curvearrowright \mu^{-1}(\phi)$ is free and proper, then there is a unique closed $(k+1)$ -form ω_ϕ on M_ϕ such that $\pi^*\omega_\phi = i^*\omega$.

$$\begin{array}{ccc} \text{restrict to } \{\mu = \phi\} & \begin{array}{c} i^*\omega \\ \pi^*\omega_\phi \end{array} & \begin{array}{c} \mu^{-1}(\phi) \xrightarrow{i} M \\ \downarrow \pi \\ M_\phi \end{array} \\ \text{quotient by } G_\phi & & \end{array}$$

Multisymplectic reduction — proof outline



- 1 Apply the *Action Descent Lemma* to $G_\phi \curvearrowright \mu^{-1}(\phi)$ and $i^*\omega$.

$$\begin{array}{ccc} i^*\omega & \mu^{-1}(\phi) & \\ & \downarrow \pi & \\ \omega_\phi & M_\phi & \end{array}$$

- 2 Use *Linear Multisymplectic Reduction* to conclude that ω_ϕ is nondegenerate.

Section 5

L_∞ -reduction

Reduction of states vs. reduction of observables

	<i>lower</i>	<i>higher</i>
<i>states</i> (geometric)	symplectic reduction	multisymplectic reduction
<i>observables</i> (algebraic)	Śniatycki–Weinstein, Dirac, Arms–Cushman–Gotay reduction	L_∞ reduction

Śniatycki–Weinstein reduction

Let (M, ω, G, μ) be a symplectic Hamiltonian G -space.

Definition (momentum ideal)

The *momentum ideal* is the associative ideal $I_\mu \subseteq C^\infty(M)$ generated by the momenta μ_ξ for any $\xi \in \mathfrak{g}$. Namely

$$I_\mu = \left\langle \mu_\xi \right\rangle_{\xi \in \mathfrak{g}}^{\text{asso.}} = \left\{ \sum_{i=1}^n f_i \mu_{\xi_i} \mid f_i \in C^\infty(M), \xi_i \in \mathfrak{g}, 1 \leq i \leq n \right\}$$

Definition

The *Śniatycki–Weinstein reduction* is the Poisson algebra

$$(C^\infty(M)/I_\mu)^G.$$

L_∞ -reduction: main construction

Suppose

- (M, ω) is a k -plectic manifold,
- $\mathfrak{g} \curvearrowright M$ is tangent to $N \subseteq M$ and preserves ω along N .

We introduce spaces

- $\text{Ham}_\infty(M, \omega)_{[N]}$ – observables **reducible** along N ,
- $I_{\text{Ham}_\infty}(N)$ – observables **vanishing** along N .

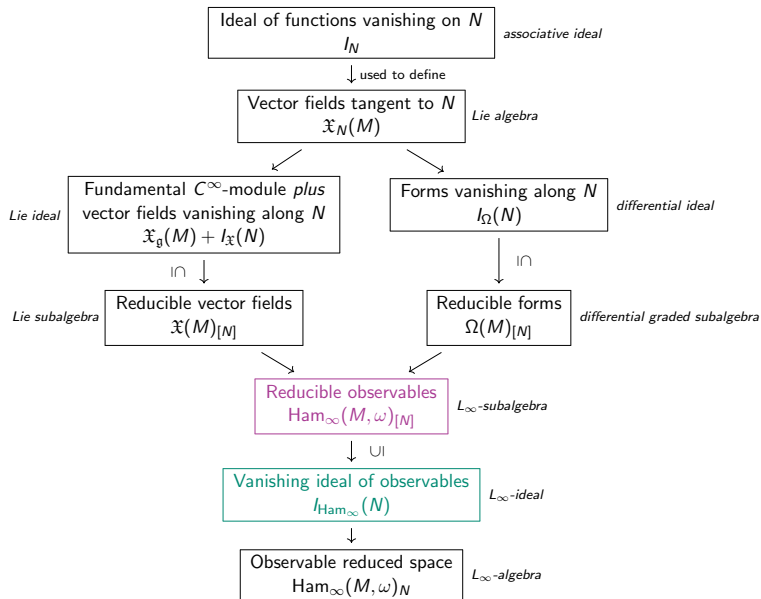
Definition (B.–Miti–Ryvkin)

The *reduction* of $\text{Ham}_\infty(M, \omega)$ with respect to $\mathfrak{g} \curvearrowright (N \subseteq M)$ is the L_∞ -algebra

$$\text{Ham}_\infty(M, \omega)_N = \frac{\text{Ham}_\infty(M, \omega)_{[N]}}{I_{\text{Ham}_\infty}(N)}.$$

In particular, the quotient inherits a **natural L_∞ -structure**.

Auxiliary constructions



Reduction with respect to a moment map

Specializing to the setting of Hamiltonian actions and moment maps yields:

Definition (B.–Miti–Ryvkin)

The *reduction* of $\text{Ham}_\infty(M, \omega)$ with respect to a

- Hamiltonian action $\mathfrak{g} \curvearrowright (M, \omega)$, and
- moment map $\mu \in \Omega(M, \mathfrak{g}^*)$,

is defined to be the reduction of $\text{Ham}_\infty(M, \omega)$ with respect to $\mathfrak{g} \curvearrowright (\mu^{-1}(0) \subseteq M)$.

It is always the case that $\mathfrak{g} \curvearrowright M$ is tangent to $\mu^{-1}(0) \subseteq M$ and preserves ω along $\mu^{-1}(0)$.

Comparison with geometric reduction

Let (M_N, ω_N) be the geometric reduction of $\mathfrak{g} \curvearrowright (M, \omega)$ along N .

Theorem (B.–Miti–Ryvkin)

The geometric reduction map

$$\begin{aligned} r_N : \mathrm{Ham}_\infty(M, \omega)_{[N]} &\rightarrow \mathrm{Ham}_\infty(M_N, \omega_N) \\ (X, \alpha) &\mapsto (X_N, \alpha_N) \\ \alpha &\mapsto \alpha_N \end{aligned}$$

is a strict L_∞ -morphism with kernel $I_{\mathrm{Ham}_\infty}(N)$. In particular, there is a natural inclusion of L_∞ -algebras

$$\mathrm{Ham}_\infty(M, \omega)_N = \frac{\mathrm{Ham}_\infty(M, \omega)_{[N]}}{I_{\mathrm{Ham}_\infty}(N)} \xhookrightarrow{\bar{r}_N} \mathrm{Ham}_\infty(M_N, \omega_N).$$

The symplectic setting

Suppose

- $G \curvearrowright (M, \omega)$ is a symplectic Hamiltonian action,
- $0 \in \mathfrak{g}^*$ is a regular value of the moment map $\mu : M \rightarrow \mathfrak{g}^*$,
- $G \curvearrowright M$ is free and proper.

Theorem (B.–Miti–Ryvkin)

The L_∞ , Śniatycki–Weinstein, Dirac, and Arms–Cushman–Gotay reductions are equal to each other.

Moreover, each is isomorphic to the Poisson algebra $C^\infty(M_0, \omega_0)$ of smooth functions on the Marsden–Weinstein reduced space.

Under broad conditions, the 1-plectic L_∞ reduced space inherits a **natural Poisson algebra structure**.

Section 6

Outlook

physics	mathematics	
classical fields	multisymplectic manifolds	✓
quantum-field symmetries	L_∞ -algebras of observables	✓
quantum fields	higher quantizations	?

Thank you!